Aaryan Lath

Aerodynamics and Systems Engineer | (312) 838-5669 | aaryanlath05@gmail.com
aaryanlath.com | linkedin.com/in/aaryan-lath

EDUCATION

Purdue University — **West Lafayette, IN**
Bachelor of Science in Aeronautical and Astronautical Engineering *(GPA: 3.74/4.0)* — *Expected May 2026*
Specialization in Aerodynamics and Systems Design
Honors: Dean's List and Semester Honors for **every** semester, **AAE Ambassador**

SKILLS

Core Tools: CAD, NX, CREO, Teamcenter, Ansys, Product Design, JIRA, Fluid Mechanics, CAM, SAP
Programming & Analysis: Python, MATLAB, Trade Studies, CFD, FEA, Financial Analysis, HTML, Java
Soft-skills: Analytical, Communicator, Detail-oriented, Fast-paced learner, Inquisitive
Awards: Accounting Case Competition 3rd place, 2nd place in Nationals of World Robot Olympiad Junior High

PROFESSIONAL EXPERIENCE

Zucrow Laboratories (High Speed Compressors Lab) — **West Lafayette, IN**
Undergraduate Research Assistant — *Aug 2025 – Present*
- Researching into the inlet vortex distortion phenomena and their effects on turbomachinery performance.
- Helped assemble and calibrate the test cell's fan-rig using laser alignment, reducing time taken by **15%**.

Siemens USA — **Grand Prairie, TX**
Systems Engineering Intern — *May 2025 – Aug 2025*
- Designed custom enclosures for panelboards using **CREO** and executed ECNs in **SAP**.
- Streamlined **100+** switchboard neutral assemblies configurations by resolving edge case designs.
- Automated BOM generation with Python, cutting manual processing time by **~30%** for **10+** orders/day.

Purdue School of Aeronautics and Astronautics — **West Lafayette, IN**
Undergraduate Teaching Assistant — *Jan 2025 – Present*
- Led study sessions to teach **50+** students, core **AAE 251** (Aircraft and Spacecraft Design) course material.
- Provided personalized support to students with concepts and **MATLAB** troubleshooting.
- Guided students in understanding key design principles for the aircraft and spacecraft design project.

Resilient Extraterrestrial Habitat Engineering — **West Lafayette, IN**
Undergraduate Research Assistant - Systems Engineer — *May 2024 – May 2025*
- Analyzed **18** potential disruptions scenarios in a lunar habitat through conducting **Simulink** simulations.
- Performed trade studies to evaluate safety controls, optimizing habitat resilience while minimizing costs through **Equivalent System Mass** analysis.
- Executed **FEA** simulations and tests to validate **vibration isolation** devices under various conditions.

PROJECTS AND INVOLVEMENTS

SAE Aero Design — **West Lafayette, IN**
Chief Engineer — *May 2025 – Present*
- Authored a **~75 page** RC aircraft design guide to accelerate onboarding
- Documented past failures from the SAE Aero Design Regular Class to avoid recurring errors.
- Led technical development for **3** sub teams by reviewing trade studies and ensured milestones are met.

Purdue Space Program – A SEDS Chapter — **West Lafayette, IN**
Satellites Deputy Systems Director — *Feb 2024 – Nov 2024*
- Improved project progress with **~15** engineers and other sub-teams ensuring alignment with project goals.
- Formulated, reviewed, and refined **200** project and system-level requirements for Boiler Bus.
- **Manufactured** the satellite frame end plates using **CAM** and walls through **laser cutting**.

Purdue Aerial Robotics Team — **West Lafayette, IN**
Competition Airframe — *Oct 2023 – May 2024*
- Performed analysis using XFLR5 to select optimal airfoils, improving lift-to-drag efficiency by ~10%.
- Engineered aircraft parts for the AUVSI SUAS competition through **NX** and Teamcenter with **~30** engineers.
- Applied engineering solutions to resolve manufacturing issues of molds, carbon fiber, and internal structures.